

Nominal Case Morphology in Kayardild: An XFST/lexc Implementation of Case Stacking

Kevin Ueffing

3595079

Computational Morphology Seminar

Summer Term 2026

Dr. Yulia Zinova

September 26, 2026

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Abstract

This paper presents an XFST/lexc implementation of nominal case morphology in Kayardild, a critically endangered Tangkic language of northern Australia (AIATSIS, 2026) noted for its capacity for recursive case stacking. The implementation covers the core case inventory across four representative declension classes, concordial and portmanteau-based case stacking, an obligatory-continuation mechanism for consonant-final adnominal suffixes, one number-marking construction, and a stem-final sandhi rule. A key structural insight, that polysyllabic case suffixes themselves carry declension-class membership, was identified through the process of compiling and testing the grammar rather than through description alone, and is discussed as a methodological contribution in its own right. All 70 test cases, built either directly from attested forms in Evans (1995) or from independently verified patterns, are correctly generated by the resulting grammar.

Abbreviations

ABL	Ablative
ACT	Actual (tense/aspect)
ADN	Adnominal
ALL	Allative
APPR	Apprehensive
ASSOC	Associative
CONS	Consequential
DU	Dual
GEN	Genitive
INSTR	Instrumental
LOC	Locative
LOT	Quantity/plural suffix (‘a lot’)
MOBL	Modal Oblique
NOM	Nominative
OBL	Oblique
ORIG	Origin
PRIV	Privative
PROP	Proprietary
UTIL	Utilitive

Glosses follow the Leipzig Glossing Rules: ‘-’ marks a morpheme boundary, ‘?’ separates multiple meanings expressed by a single non-segmentable form (e.g.

child.NOM), and ‘:’ marks a portmanteau suffix realizing two categories at once (e.g. *LOC:MOBL*).

1 Introduction

The Kayardild language, spoken in northern Australia, is noted among linguists for its unusual case system, most famously its capacity for recursive case stacking. This project was initially motivated by a popular meme circulated on the `r/linguisticshumor` subreddit, which dramatizes Kayardild as one of the hardest languages to parse due to its extreme grammatical case escalation. Consulting the descriptive literature (Evans, 1995; Round, 2013) confirmed that this phenomenon, known as *case-stacking* or *Suffixaufnahme* (Dench and Evans, 1988), is indeed a central and well-documented feature of the language’s grammar. From a computational perspective, modeling such recursive morphological layering poses an interesting morphotactic and morphophonological modeling problem for finite-state architectures (Beesley and Karttunen, 2003).

Kayardild is a typologically dependent-marking, highly agglutinating language. Crucially, it exhibits no verbal agreement, relying almost entirely on its extensive inflectional case system to express both core grammatical relations and complex clausal dependencies.

This paper focuses specifically on the nominal case system of Kayardild, comprising core case marking, case stacking (both concordial and modal case), number marking, and stem-final consonant alternations, restricted to the noun-adjective subclass (see Section 4 for full scope delimitation).

The remainder of this paper is structured as follows. Section 2 provides background on the language and its speakers. Section 3 introduces the core case system and the phenomenon of case stacking. Section 4 delimits the scope of this work. Section 5 discusses the methodological choices underlying the implementation, including the decision to rely primarily on Evans (1995) as the data source. Section 6 describes the `lexc/xfst` implementation. Section 7 presents the test suite and evaluation. Section 8 discusses the results and limitations of the implementation. Section 9 summarizes the paper’s contributions.

2 Language Background

Kayardild is a non-Pama-Nyungan language belonging to the Tangkic language family, specifically forming the South Wellesley subgroup alongside the closely

related Yangkaal (Evans, 1995, p. 12). Ethno-etymologically, the name derives from *ka(ng)* (‘language’) and *yardild(a)* (‘strong’) (Evans, 1995, pp. 8–9). It was traditionally spoken on Bentinck and Sweers Islands in the Gulf of Carpentaria, Queensland. Due to extreme geographical and social isolation prior to full and sustained European contact in 1948 (Evans, 1995, pp. 39–43), Kayardild maintained an exceptionally homogeneous grammar with minimal dialectal variation or borrowing from neighboring mainland languages.

Today, Kayardild is critically endangered (AIATSIS, 2026). By the late 20th century, fieldwork observed a rapid language shift toward Mornington English and Lardil, with fluent usage becoming increasingly restricted to older generations and specific social contexts (Evans, 1995, pp. 43–47). In 2016, the Australian Census recorded only eight native speakers (Australian Bureau of Statistics, 2016).

Typologically, traditional conservative Kayardild features relatively free word order driven by discourse structure, as syntactic relationships are unambiguously coded through its rich inflectional morphology. Unlike many Australian languages that exhibit ergative-absolutive alignment (Dixon, 1980), Kayardild displays a consistent nominative-accusative pattern. In younger semi-speakers, English influence has led to a fixed SVO constituent order and a reduction in object case marking. This project, however, explicitly models the conservative variety described by Evans (1995).

3 The Phenomenon: Case and Case Stacking in Kayardild

This section introduces the two layers of nominal case marking that this paper models: the core case system, which marks basic grammatical relations, and the phenomenon of case stacking, whereby multiple case suffixes can co-occur on a single nominal to encode different levels of syntactic embedding or clause-level modal information. Section 3.1 presents the core case inventory, while Section 3.2 turns to case stacking itself.

3.1 The Core Case System

Kayardild nominals inflect for an extensive system of case, encompassing twelve distinct case categories in addition to the nominative, which functions as an elsewhere case realized whenever no other relational, modal, associating, or complementizing case is assigned (Evans, 1995, p. 136). Each case carries both an adnominal function (marking relations within the noun phrase) and a relational function (marking the nominal’s role with respect to the clause).

Case	Primary function(s)
Nominative	Elsewhere case; citation form; core clausal subjects
Locative	Location; extended to time, cause, manner
Ablative	Source; past-event possession; demoted agents
Propriative	(Potential) having; instruments; intentional objects
Oblique	Non-committal relational case; comparison; kin suitability
Allative	Direction of motion; extensive/pergressive/perlative uses
Genitive	Possession (kin, land, allocation, material source)
Instrumental	Instrument of contact
Associative	Temporary co-location, possession, accompaniment
Origin	Provenance; characteristic location or time
Privative	Lack/absence; negation
Consequential	Prior contact/involvement; cause
Utilitive	Conventional/intended use

Table 1: Core case inventory of Kayardild, adapted from Evans (1995, p. 135)

Many cases additionally carry a modal function, discussed in Section 3.2. Table 1 summarizes the core cases and their primary functions.

As an illustration, the locative in (1) marks the location at which an event takes place.

- (1) ngaaka dangka-a waa-ja ngambirr-iy
 who-NOM person-NOM sing-ACT humpy-LOC
 ‘Who is singing in the humpy?’

As a further illustration, the genitive in (2) marks adnominal possession (Evans, 1995, p. 151).

- (2) dathin-karra maku-karra kularrin-da kurrka-th
 that-GEN woman-GEN brother-NOM take-ACT
 ‘That woman’s brother took it.’

3.2 Case Stacking

Case stacking refers to the co-occurrence of multiple case suffixes on a single nominal, each suffix corresponding to a distinct rank in the template proposed by Evans (1995, p. 122): a stem may take an arbitrary number of adnominal or number suffixes, followed by up to two relational case suffixes, followed by up to two modal case suffixes, optionally followed by a mutually exclusive

associating or complementizing case suffix. This section illustrates the two most prominent stacking mechanisms (Dench and Evans, 1988): concordial stacking, whereby an already case-marked nominal receives an additional case suffix reflecting the case of a superordinate constituent, and modal case, whereby the outermost case on an argument encodes tense, mood, or aspect for the entire clause.

Concordial stacking. In concordial stacking, a nominal already inflected for one case (typically marking a possessor or an embedded relation) receives a further case suffix agreeing with the case assigned to the phrase in which it is embedded. In example (3), the genitive-marked possessor additionally carries the locative case of the whole possessive phrase (Evans, 1995, p. 152).

- (3) nal-karran-ji
 head-GEN-LOC
 ‘around (my) head’

A second, more complex example shows that this recursion is not limited to a single additional layer: adnominal and number suffixes, including LOT (‘a lot’, a plural/quantity suffix discussed further in Section 6), may co-occur multiple times and in either relative order (Evans, 1995, p. 123), as shown in example (4).

- (4) maku-yarr-nurru-naba-walad
 woman-DU-ASSOC-ABL-LOT
 ‘the many belonging to (those) having two wives’¹

In this example, the Associative and Ablative suffixes function adnominally (marking relations within the possessor phrase) rather than relationally, and therefore occupy the recursive adnominal rank alongside the Dual and LOT suffixes, not the relational rank.

Modal case. Beyond marking relations within or below the clause, the outermost case on a nominal argument may encode tense, mood, or aspect for the entire proposition, which is a function typically associated with verbal inflection in other languages. In example (5), the modal oblique on the object marks the apprehensive modality of the clause as a whole.

¹The reduced forms *-yarr-* (from the full Dual *-kiyarrng-*) and the retained *-naba-* (rather than the word-final-reduced *-na*) are consistent with independently attested alternation patterns (Sections 5, 6.4) but are not themselves implemented, since Dual number marking falls outside the scope of this work (Section 4).

- (5) kunawuna bilarri-nyarra nguku-ntha wuruman-kurrk
 child.NOM spill-APPR water-MOBL billy-(ADN)LOC:MOBL
 ‘The kid might spill the water in the billy.’

Here, the suffix *-kurrk* realizes a portmanteau of the adnominal locative and the modal oblique, reflecting a sequence restriction whereby the locative cannot otherwise be followed by any inflection but the oblique (Evans, 1995, p. 139).

4 Scope and Delimitation

Subclass restriction. Evans notes that the case system is essentially identical across all nominal subclasses, including noun-adjectives, pronouns, locationals, and manner and time nominals, differing only in minor restrictions on adnominal and relational inflections for the latter three subclasses (Evans, 1995, p. 122). This implementation is accordingly restricted to the noun-adjective subclass as a representative case; derivational, compounding, and reduplicational processes specific to this subclass are likewise excluded, as they constitute word formation rather than inflection.

Stacking depth. The rank template given in Evans (1995, p. 122) permits an unbounded number of adnominal and number suffixes, up to two relational case suffixes, and up to two modal case suffixes. While the adnominal/number rank is illustrated with recursion up to four suffixes deep in example (4), this implementation limits recursion at this rank to two occurrences. This restriction keeps the rule count within the scope of the assignment while still exhibiting genuine stacking behavior, and reflects the fact that examples with more than two adnominal suffixes are comparatively rare in the source grammar.

Exclusions. The following are explicitly excluded from this implementation:

- **Verbal case** (Evans, 1995, pp. 163–183), whereby certain case-marked nominals are converted into morphologically verb-like forms. This system interacts closely with verbal tense-aspect-mood marking and falls outside the nominal focus of this work; Round (2013) offers an alternative analysis of this interaction, discussed further in Section 5.
- **Pronouns, locationals, manner and time nominals**, which largely share the core case system but exhibit subclass-specific restrictions and derivational processes not covered here.

- **The associating and complementizing functions of the oblique case** (Rank 4 in the template of Evans (1995, p. 122)), which interact with clausal subordination and nominalized clause structure and are therefore excluded alongside subordinate clause constructions more generally. Note that this exclusion concerns a specific *functional use* of the oblique case at the outermost rank, not the oblique case itself, which remains part of the core case inventory (Table 1).

The resulting scope comprises the nominative and twelve further core cases, concordial stacking at the adnominal rank (limited to two occurrences), and modal case at the relational and modal ranks, together with the stem-final consonant alternations conditioned by declension class (Section 6).

5 Methodology

5.1 Formalism and Notation

Morphotactics (the inventory of stems, suffixes, and their licensed combinations) is implemented in *lexc*, while morphophonological alternations are implemented as *xfst* replace rules and composed with the *lexc*-compiled lexicon (Beesley and Karttunen, 2003). *twolc*, an alternative two-level rule formalism (Koskeniemi, 1983; Beesley and Karttunen, 2003), is not used here, as the alternations targeted in this implementation (Section 6) are straightforwardly expressible as ordered context-sensitive replacements within *xfst* itself.

Grammatical categories are represented as multicharacter symbols prefixed with `+` (e.g. `+NOM`, `+LOC`). This avoids the need to escape reserved characters such as `<` and `>` and keeps analyses such as `yakuri+NOM` readable without additional markup.

Stems and suffixes are transcribed in Kayardild’s practical ASCII orthography (e.g. *rt*, *rl*, *ng*, *ny* for retroflex and palatal segments) rather than in IPA, consistent with the source grammar’s own citation forms. This also avoids encoding issues in plain-text *lexc* files, since not all *xfst* implementations reliably handle non-ASCII Unicode input.

5.2 Evans (1995) versus Round (2013)

Although this project was initially motivated by Round’s (2013) reanalysis of Kayardild’s case and TAM systems (see Section 1), the implementation relies primarily on Evans (1995) as its data source. This choice reflects the scope of the project: Round’s central innovation, a unified treatment of verbal and nominal TAM as “formal case”, realized differently on thematic

versus athematic stems, bears on the interaction between nominal and verbal morphology, which falls outside the nominal-only scope defined in Section 4. Within the nominal domain proper, Evans’ description already provides an explicit rank template (Section 3) and a fully specified declension-class system (Section 6), sufficient to drive the implementation without requiring Round’s (2013) reanalysis.

5.3 Design Decisions

Declension class selection. Evans (1995) distinguishes nine declension classes (1, 2, 3a, 3b, 4a, 4b, 5a, 5b, 6), conditioned by the phonological shape of the stem-final segment. Implementing all nine would have required approximately 117 continuation-class lines for Rank 1 alone, exceeding the intended scope. This implementation therefore covers four representative classes, one per major stem-final category: vowel-final in *i* (Declension 1) and *u* (Declension 2), vowel-final in *a* (Declension 3a), and consonant-final (Declension 4a).

Within each vowel-final category, the simpler of the available sub-classes was chosen: Declension 3a rather than 3b, since 3a covers the more frequent disyllabic case and avoids additional prosodic truncation effects specific to longer 3b stems (Evans, 1995, Section 2.3). Declension 4a rather than 4b was chosen because its stem-final segments (*r*, *l*, *th*, *j*) directly motivate the stem-final sandhi alternations that are the focus of Section 6, making the choice substantively rather than arbitrarily motivated.

Suffixes as declension-class-bearing stems. A recurring design principle, discovered during implementation rather than anticipated in advance, is that polysyllabic case suffixes themselves behave like derived stems with their own declension class membership, exactly as roots do. Evans (1995, p. 127) notes this explicitly: the Propriative suffix *-kuru*, being *u*-final, selects the same following allomorphs as a *u*-final root such as *maku* ‘woman’. The same holds for the Genitive’s full form *-karran-* and the Origin suffix’s full form *-wan-*, both *n*-final: any suffix following either of them, rather than following the declension class of the original root, takes the allomorph appropriate to *n*-final stems. This is implemented as a dedicated continuation lexicon (**Rank2-nfinal**) shared by every suffix that is itself *n*-final, regardless of which root it attaches to.

This principle was initially overlooked. An early illustrative example for concordial stacking (Section 3.2) was constructed by analogy rather than drawn from the source text, and assumed that the Locative suffix following the Genitive would take its ordinary root-conditioned form. Compiling the

lexc grammar and testing this example surfaced a mismatch with the independently attested form *nal-karran-ji* (Evans, 1995, p. 152), which prompted the correction described above. This is reported here not merely as an implementation detail but as a methodological point: the compiled grammar itself functioned as a consistency check against the hand-built analysis, catching an error that closer reading alone had not surfaced.

The Origin suffix’s full form, cited here as *-wan-* following its form in Evans’ paradigm table (Figure 4-2), also occurs with a long vowel as *-waan-* in Evans’ prose (e.g. *ngimi-waan-da*, p. 137); Evans himself heads the relevant section as “ORIGin *-wa(a)n-*” (Evans, 1995, Section 4.3.11), reflecting the same variability. Consistent with the citation-form principle above, the short form is adopted throughout.

Handling variant and optional forms. Several suffixes admit more than one grammatical surface form without a fully general conditioning rule. The Propriative’s word-final truncation, for instance, is reported by Evans (1995, Section 4.3.5.1) to vary by speaker age, sex, and style rather than by phonological position. Rather than adjudicate between such variants, this implementation consistently adopts the form Evans himself uses as the citation form in his paradigm tables (Figure 4-2), even where an alternative reduced form is independently grammatical. The same principle applies to the Nominative augment (*-ya*, *-wa*, *-a*): Evans (1995, p. 138) analyzes this added segment as only partly phonologically motivated, likely originating as a discourse particle rather than a fully regular suffix, and it is likewise implemented in its cited form even where the source describes it as optional. This principle, deferring to the source’s own citation form wherever a choice between non-conditioned variants is otherwise required, is applied uniformly rather than being decided case by case.

Obligatory continuations. Two suffixes, the Genitive’s full form and the Origin suffix, cannot occur at the end of a word in their bare form: both are consonant-final, and Evans (1995, p. 137) states that a following Nominative is required whenever no other case is present (e.g. *ngimi-waan-da* ‘night-ORIG-NOM’, not **ngimi-waan*). This is implemented structurally rather than through a filter: the lexc entries for these suffixes provide no path directly to the word boundary, forcing every analysis to continue into **Rank2-nfinal**, which supplies the closing Nominative (or an alternative continuation, where attested). An analysis lacking a licensed continuation, such as **+ORIG** alone, is consequently rejected by the grammar itself, without any additional well-formedness check.

Distinguishing relational and modal oblique. The Oblique case realizes two distinct functions in Evans’ analysis: a plain relational function, and a modal function that, when combined with a preceding Locative, is realized as the portmanteau *-kurrka* rather than as concatenated Locative and Oblique suffixes, a sequence restriction Evans states explicitly (Evans, 1995, pp. 129–130). Since these two functions have different morphological consequences, they are kept distinct at the tag level: **+OBL** for the relational function, and a separate **+MOBL** for the modal function, mirroring the **LOC:MOBL** notation Evans himself uses when glossing the portmanteau (Evans, 1995, p. 139, example 4-18).

Verification methodology. Every suffix form and every example used to justify a structural decision was checked directly against Evans (1995) before being implemented, rather than being constructed by analogy or recalled from memory. Where a plausible generalization could not be independently attested, such as the irregular alternation *nal* to *nat* before the Utilitive (Section 6), it was implemented as a lexically restricted exception rather than generalized into a rule, and is reported as such rather than as an independently confirmed phonological process.

In addition to the systematic positive coverage described in Section 7, key negative cases were verified interactively during development via **apply down**: for example, **nal+UTIL**, a bare **+ORIG** (with no licensed continuation), and a bare **+LOT** were confirmed to be correctly rejected by the compiled grammar. These checks were not, however, integrated into the automated test suite (Section 7), which verifies only positive coverage.

6 Implementation

6.1 Architecture

The implementation consists of two core files. **kayardild.txt** contains the morphotactic lexicon in lexc format: stems, declension classes, and the continuation structure governing case, stacking, and number marking. **sandhi.txt** contains the phonological replace rule described in Section 6.4. A single script, **kayardild.xfst**, loads both, composes them into a single transducer, and applies every test case (Section 7) in one step, writing the results to **kayardild_output.txt**:

```

clear stack;
read lexc < kayardild.txt
define Lexicon
source sandhi.txt
regex Lexicon .o. ClusterSimplification;

apply down < kayardild_inputs.txt > kayardild_output.txt

```

The resulting output is compared against `kayardild_goals.txt`, a reference file containing the expected surface forms for every test case (Section 7).

Throughout this section, example forms are hyphenated to indicate morpheme boundaries for expository clarity (e.g. *nat-marra*, *wangalk-*); the surface forms actually generated by the grammar contain no such boundary markers (e.g. *natmarra*, *wangalnguni*), as shown in the test output discussed in Section 7.

6.2 Lexicon Structure

The lexicon is organized as `Root` $\rightarrow$ `Nouns` $\rightarrow$ one continuation lexicon per declension class (`Decl1`, `Decl2`, `Decl3a`, `Decl4a`), following the declension selection described in Section 5. Each declension lexicon lists twelve of the thirteen core cases (all but the `Origin`, see below) with their class-specific allomorphs:

```

LEXICON Decl1
+NOM:ya # ;
+LOC:ya # ;
+ABL:na # ;
+PROP:wuru # ;
+OBL:nja # ;
+ALL:r # ;
+GEN:karra # ;
+ASSOC:nurru # ;
+PRIV:warri # ;
+CONS:ngarrba # ;
+INSTR:nguni # ;
+UTIL:marra # ;

```

The `Origin` case is deliberately omitted from this listing. Unlike the cases shown above, it cannot terminate a word on its own and is instead routed through the stacking mechanism described in Section 6.3, alongside the Genitive’s obligatory continuation. Declensions 2, 3a, and 4a follow the same overall structure, differing only in their class-specific allomorphs.

Four stems, one per declension class, instantiate the paradigm: *yakuri* ‘fish’ (Declension 1), *maku* ‘woman’ (Declension 2), *mala* ‘sea’ (Declension 3a), and *nal* ‘head’ (Declension 4a). A fifth stem, *wangalk-* ‘boomerang’, is added separately to illustrate the sandhi rule (Section 6.4).

6.3 Case Stacking

Concordial stacking. The Genitive’s full form *-karran-* is licensed by two alternative continuations, reflecting the two attested outcomes depending on what, if anything, follows:

```
+GEN:karra # ; ! word-final, reduced form
+GEN:karran Rank2-nfinal ; ! full form, continues
```

Rank2-nfinal supplies the allomorphs appropriate to an *n*-final stem (Section 5), independent of the declension class of the original root:

```
LEXICON Rank2-nfinal
+NOM:da # ; ! e.g. nal-wan-da (Origin + Nominative)
+LOC:ji # ; ! e.g. nal-karran-ji (Genitive + Locative)
```

The Origin suffix, whose full form *-wan-* is likewise *n*-final and cannot occur word-finally on its own (Section 5), is routed through the same continuation lexicon:

```
+ORIG:wan Rank2-nfinal ; ! obligatory continuation
```

The LOC+MOBL portmanteau. The Locative admits two continuations: a terminal path realizing the plain relational Locative, and a path via an empty suffix into *LOC-Modal*, which supplies the portmanteau realizing Locative plus modal Oblique as a single suffix:

```
+LOC:ya # ; ! plain Locative
+LOC:0 LOC-Modal ; ! begins the portmanteau path

LEXICON LOC-Modal
+MOBL:kurrk # ;
```

Unlike the Genitive-conditioned continuation above, this portmanteau form is identical across all declension classes, since *-kurrka* is a fixed, non-declension-conditioned form (Evans, 1995, pp. 129–130); a single *LOC-Modal* lexicon therefore serves all four declensions.

6.4 Lexical and Phonological Refinements

A lexicalized exception. The stem *nal* ‘head’ forms all cases regularly except the Utilitive, which Evans’ paradigm (Figure 4-2) gives as *nat-marra* rather than the expected **nal-marra*.² Since no general rule accounting for this alternation could be located, it is implemented as a lexically restricted suppletive stem rather than as a phonological rule: *nal* continues into Decl14a, which excludes the Utilitive entirely, while a separate stem *nat* licenses only the Utilitive:

```
LEXICON Nouns
nal Decl14a ;
nat Decl14a-UTIL-irregular ;

LEXICON Decl14a-UTIL-irregular
+UTIL:marra # ;
```

The analysis *nal*+UTIL is consequently not accepted by the grammar; only *nat*+UTIL is.

Number: LOT. Of the five number-related suffixes described in Evans (1995, Section 4.5), only LOT is implemented, for the reasons given in Section 8. Its base form *-walath-* is *th*-final and delaminalizes to *-walad-* before the Nominative (Evans, 1995, Section 2.5.3); since only the LOT+Nominative combination is implemented, the delaminalized form is stored directly rather than derived by a separate rule:

```
+LOT:walad LOT-Cont ;

LEXICON LOT-Cont
+NOM:a # ;
```

Sandhi: cluster simplification. Evans (1995, p. 76) describes a general rule of cluster simplification. Phonotactically illicit consonant clusters arising at a morpheme boundary are repaired by deleting the penultimate segment of the cluster. For a stem ending in the cluster *-lk* (as in *wangalk-* ‘boomerang’), followed by a suffix beginning in a consonant, this deletes the *k*:

²A systematic search for an explanation was conducted across Evans (1995, Sections 2.5.1–2.5.6) (nasal assimilation, lateralization, delaminalization, feature blends, and cluster simplification, respectively) and, since Round (2013) offers an independent morphophonological reanalysis of the language, across the relevant sections of Round (2013) as well. No rule predicting this specific alternation was found in either source.

```
define ClusterSimplification [ k -> 0 || 1 _ [ n | w ] ] ;
```

Note that only grammatical tags are declared as multicharacter symbols (Section 5); phonological digraphs such as *ng* are not, so the rule correctly matches the literal segment *n* at the start of a suffix such as *-nguni*.

This is implemented as an xfst replace rule rather than as a lexical listing, since, unlike the suppletive exception above, it is explicitly stated as a general phonological process in the source and is independent of any single stem. The rule is verified against the two directly attested outcomes, *wangalk-* + *-nguni* (Instrumental) → *wangalnguni* and *wangalk-* + *-warri* (Privative) → *wangalwarri* (Evans, 1995, p. 76, example 2-18); its behavior on two further, untested combinations is discussed in Section 8.

7 Testing

7.1 Test Suite Design

The test suite consists of 70 test cases, each specifying an analysis (e.g. **nal+GEN+LOC**), the surface form the grammar is expected to produce for it, and a status label distinguishing two kinds of evidence. Cases labeled *attested* reproduce a form that appears verbatim in Evans (1995), either in the case paradigm (Figure 4-2) or in a specific numbered example; cases labeled *analogous* follow a pattern independently attested elsewhere in the grammar but are not themselves drawn from a directly cited form for the particular stem in question. For instance, **nal+GEN+LOC** → *nalkarranji* is attested directly (Evans, 1995, p. 152, example 4-68), while the corresponding form for *yakuri* is analogous: it follows the same independently verified rule (Section 6.3) but is not itself quoted in the source. This distinction is recorded for every test case so that the strength of evidence behind each result remains transparent rather than being implicitly conflated.

7.2 Coverage

The suite is organized to test every construction described in Sections 3 and 6 at least once, and, where the construction is declension-independent, across each of the four implemented declension classes:

- all thirteen core cases (Rank 1), for each of *yakuri*, *maku*, *mala*, and *nal* (with *nal+UTIL* deliberately excluded and *nat+UTIL* tested in its place, per the lexicalized exception in Section 6);
- concordial stacking (+GEN+LOC) and the obligatory-continuation cases (+ORIG+NOM, +ORIG+LOC), for all four stems;
- the LOC+MOBL portmanteau, for all four stems;
- the LOT+Nominative number construction, for all four stems;
- the cluster simplification rule, for *wangalk-* with the two directly attested suffixes (Instrumental, Privative).

Combinations that are not licensed by the grammar, such as a bare +ORIG with no continuation or *nal+UTIL*, are excluded from the suite rather than included as expected failures, since the suite is intended to verify correct positive coverage rather than to enumerate the grammar’s rejections. Two forms that the cluster simplification rule predicts but that are not independently attested (*wangalnurru*, *wangalngarrba*; Section 8) are likewise excluded from the suite, consistent with the attested/analogous distinction: including them without a citation would blur the line between verified and predicted output.

7.3 Execution and Results

Execution. Testing is performed by `kayardild.xfst` (Section 6), which composes the lexicon with the sandhi rule and applies every analysis listed in `kayardild_inputs.txt`, writing the resulting surface forms to `kayardild_output.txt`. A Python script invokes this xfst script as a subprocess, then compares `kayardild_output.txt` against a reference file, `kayardild_goals.txt`, using Python’s `diff` module to report any line-level discrepancies between the two.

Results. All 70 test cases pass: the generated output is identical to the reference file, with no discrepancies reported.

8 Discussion

8.1 What Worked Well

The declension-class architecture generalized cleanly to case stacking: once the observation that polysyllabic suffixes themselves carry declension-class membership (Section 5) was identified, a single continuation lexicon (**Rank2-nfinal**) correctly handled both the Genitive’s and the Origin’s obligatory or optional continuations without stem-specific duplication. The compilation process itself also proved useful as a consistency check: an early example constructed by analogy rather than drawn from the source text was shown to be incorrect only once the compiled grammar was tested against it, at which point the correct, independently attested form (*nal-karran-ji*) was substituted (Section 5). This suggests that building and testing a computational implementation alongside a grammatical description can surface inconsistencies that close reading alone does not.

8.2 What Was Harder Than Expected

Several suffixes admit variation that is not conditioned by phonological position but by sociolinguistic factors such as speaker age, sex, or style (e.g. the Propriative’s word-final truncation, Evans, 1995, Section 4.3.5.1) or by an unclear historical origin (the Nominative augment, Evans, 1995, p. 138). To maintain a deterministic baseline, avoid non-deterministic overgeneration, and keep the compiled lexicon aligned with the citation forms used in Evans (1995) (Section 5), resolving these cases required adopting a single canonical variant for each suffix, rather than a linguistic explanation for the variation itself; this is a necessary simplification relative to the variation actually described in the source.

One alternation resisted explanation entirely: the stem *nal* ‘head’ forms the Utilitive as *nat-marra* rather than the expected **nal-marra* (Section 6). As noted there, a search covering both Evans’ (1995) morphophonemic rules and Round’s (2013) independent reanalysis did not surface a rule predicting this alternation. It is implemented as a lexicalized exception rather than as a rule, and is reported here as an open point rather than as a resolved one.

A cleaner architecture would abstract over this alternation at the lexical level entirely, accepting a single underlying lexeme **nal** for all cases including the Utilitive (**nal+UTIL**) and deriving *natmarra* via a stem-and-suffix-specific replacement rule, rather than introducing a second surface stem. This was not pursued here, since no independently attested phonological rule could justify such a replacement (Section 6), and an ad-hoc rule restricted to this one

lexeme would not meaningfully differ in generality from the suppletive-stem solution adopted; it is noted here as a possible refinement rather than a currently motivated one.

8.3 Material Knowingly Not Covered

Consistent with the scope defined in Section 4, several categories of data were deliberately excluded:

- Five of Evans’ (1995) nine declension classes (3b, 4b, 5a, 5b, 6) are not implemented; the four classes covered were chosen to represent each major stem-final category while keeping the implementation within scope (Section 5).
- Of the five number-related suffixes described in Evans (1995, Section 4.5), only LOT is implemented. Dual involves declension-conditioned allomorphy parallel to the Locative that was not independently verified within the scope of this project; PLENTY involves nasal-conditioned allomorphy ($w \rightarrow m$ after nasal-final stems); EVERY has a clitic-like origin with no attested following inflections. All three, together with ANOTHER, were judged to fall outside the intended scope.
- The associating and complementizing functions of the Oblique (Rank 4 in Evans’ template) are excluded, as they interact with subordinate clause structure, itself outside the nominal scope of this work (Section 4).
- The cluster simplification rule (Section 6.4) is verified against two directly attested outcomes (*wangalnguni*, *wangalwarri*). Applied generally, the same rule also predicts *wangalnurru* and *wangalngarrba*; these are not independently attested in the consulted excerpts and are accordingly not included in the test suite as verified forms.

More generally, the test suite verifies only positive coverage; while key negative cases (e.g. bare +ORIG, **nal**+UTIL, bare +LOT) were confirmed interactively during development to be correctly rejected (Section 5), these checks are not part of the automated suite. Nor does the current evaluation exploit the bidirectionality of the transducer via **apply up** to test parsing from surface form back to analysis. Both would strengthen the evaluation and are left as future work.

8.4 Possible Extensions

A natural extension beyond the material excluded above would be to integrate Round’s (2013) unified treatment of nominal and verbal TAM as thematic and athematic realizations of a single underlying system (Section 1). Since this project’s scope is limited to nominal morphology, this reanalysis was not required for the constructions implemented here, but it would become directly relevant if the scope were extended to include the Verbal Case system (Evans, 1995, Section 4.4), which this work excludes.

9 Conclusion

This paper presented an XFST/lexc implementation of nominal case morphology in Kayardild, formalizing a representative portion of the language’s typologically renowned case-stacking system (Dench and Evans, 1988) across four declension classes. The implementation covers the core case inventory, concordial and portmanteau-based case stacking, an obligatory-continuation mechanism for consonant-final adnominal suffixes, the LOT number construction, and a phonotactic cluster simplification rule. All 70 test cases, built either directly from attested forms in Evans (1995) or from independently verified patterns, are correctly derived by the compiled grammar.

Beyond capturing the target phenomena, the implementation demonstrates a clear methodological benefit: key structural insights, most notably that polysyllabic case suffixes carry their own declension-class membership, were surfaced directly through the iterative compilation and testing process. Unresolved alternations, such as the irregular Utilitive form *nat-marra*, were transparently documented as lexicalized exceptions rather than obscured by unsupported rules.

The scope deliberately excludes verbal morphology, five of Evans’ (1995) nine declension classes, four of five number-related suffixes, and the associating and complementizing functions of the Oblique; each exclusion is motivated in Section 4 and revisited in Section 8. Extending the automated test suite with the negative cases already verified interactively during development, together with bidirectional (`apply up`) parsing evaluation, constitutes the most direct avenue for future work.

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